Bingxi (Paul) Jiang

(949) 344-0799 | pauljiang817@gmail.com | www.linkedin.com/in/bingxi-jiang | github.com/Bingxi-Jiang

EDUCATION

University of California, Irvine Expected June 2027
Bachelor's Degree, Computer Science | GPA: 3.835/4.0 Irvine, CA

TECHNICAL SKILLS

Languages Python, JavaScript, TypeScript, C++, Java, SQL, Go, Shell

Web & APIs React, Next.js, Node.js, Express, Spring Boot, FastAPI, HTML/CSS, REST, GraphQL, HTTP, WebSockets, OAuth 2.0, JWT

Databases & Messaging PostgreSQL, MySQL, MongoDB, Redis, Kafka, Elasticsearch, BullMQ

Cloud & DevOps Linux, AWS, GCP, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Sentry, Prometheus

Engineering Data Structures & Algorithms, Object-Oriented Programming, Distributed Systems, System Design, Operating Systems

Testing Playwright, Jest, pytest, Postman; Unit, Integration, E2E, Regression, Mocking, Load Testing, TDD

AI Development Cursor, Claude Code, GitHub Copilot, MCP, LangChain, RAG

PROFESSIONAL EXPERIENCE

Software Engineer Intern June 2026 - September 2026
Dassault Systèmes San Diego, CA

- Built a closed-loop LLM pipeline using Playwright MCP to automate the conversion of legacy Geb workflows into TypeScript.
- Architected a modular TypeScript framework with reusable API, nested-frame, and AG Grid, enabling faster development and validation.
- Cut a 23-step enterprise workflow from 15–20 min to 3–4 min while expanding validation checkpoints from 6 to 23.
- Enabled idempotent reruns with isolated BrowserContexts, storageState, and persisted workflow IDs across user roles.
- Replaced fixed waits with response, DOM, and lifecycle-driven synchronization, retaining trace/video artifacts for diagnosis.

Software Engineer Intern April 2026 - June 2026
GTechFin Inc (B2C Fintech startup) New York, NY

- Architected a Python/Typer workflow engine and MCP service with rerunnable DAG, improving reliability and reducing process error.
- Modeled pipeline state in MySQL with transactions, indexes, and idempotent retries, improving consistency and query latency.
- Built a self-hosted cosine index for 500–5K videos after benchmarking MongoDB and Elasticsearch, accelerating search and retrieval.
- Cut video production from ~3 hours to ~20 minutes across 7 benchmarks against professional and freelance workflows.

Software Engineer Intern September 2025 - December 2025
Fresh Road Inc (B2B SaaS startup) San Jose, CA

- Built a real-time Node.js/Express voice backend streaming Twilio audio to Deepgram, LLMs, and CRM services, enabling concurrent calls.
- Orchestrated six BullMQ/Redis queues with retries, backoff, deduplication, and recovery, improving reliability and queue latency.
- Cut LLM cost 32% and p95 latency 18% via caching, compressed contexts, task-based routing, and batched/parallel inference.
- Modeled multi-tenant call state in MongoDB with compound indexes and idempotency controls, improving consistency and debugging.
- Traced HTTP, WebSocket, queue, LLM, and CRM flows with correlation IDs, Sentry, and typed failures, accelerating incident resolution.

Algorithm Engineer Intern June 2025 - September 2025
Ping An Technology Guangdong, China

- Built a six-stage Pandas/MySQL pipeline for ingestion, QC, classification, scoring, and reporting, cutting preprocessing by 30%.
- Built an async FastAPI/WebSocket inference API with job tracking and structured output, returning results within seconds.
- Optimized INT8 serving with dynamic batching, cutting p95 latency 38% and raising throughput 50% across fixed benchmarks.
- Built a 2.27M-image Docker/Pandas/MySQL pipeline with deduplication and leakage-safe splits, improving training data quality.
- Packaged prototypes in pinned Docker images and Kubernetes services adopted downstream in systems handling 10K+ daily requests.

PROJECTS

FocusTrail: AI Task Breakdown & Real-Time Monitor github.com/PST-Protocol/FocusTrail

- Built Gemini-powered task orchestration system with React and Vite to break complex tasks, improving execution efficiency by 43%.
- Built Node.js and Express backend with Server-Sent Events streaming real-time focus telemetry from a macOS osascript agent.
- Deployed full-stack productivity platform on GCP with REST APIs for breakdown, monitor sessions, stats aggregation and history.

Noted: Real-Time Multimodal Transcription & Memory System github.com/Bingxi-Jiang/Noted

- Architected real-time transcription pipeline streaming 16kHz PCM audio via WebSocket to Deepgram Nova-2, achieving <300ms latency.
- Engineered multimodal memory system combining transcripts, speaker diarization, and Gemini Vision analysis for context preservation.
- Implemented cross-session RAG with embedding similarity search over transcript chunks, enabling context-aware retrieval across folders.

AWARDS

1st Place, Streamlit AI Hack

Won 1st Place at Streamlit AI Hack by building a Snowflake Cortex app forecasting reactions to tech launches across cities.

Built the working POC on Snowflake under hackathon constraints, turning the product concept into a demo within hours.

Presented the product to Snowflake and Streamlit judges, explaining architecture, business value, tradeoffs, and Q&A